JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of CSE

"A place to learn; A chance to grow"

2025-2026(Odd Semester)

Education to Eternity

MSE-I EXAMINATION

Course: B.tech in CSE
Subject: Blockchain Technology
Max Marks: 20

Date: 07/08/2025

Year/Semester: 5^{th} Semester (3^{rd} Year)
Subject Code: CS5E001A
Duration: - 1 Hr.

Instructions to the Students:
1. **All the questions carry marks as indicated.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

Students should be able to:

CO 1: Students will be able to Understand emerging abstract models for Blockchain Technology.

CO 2: Students will be able to identify major research challenges and technical gaps existing between theory and practice in the crypto currency domain.

CO 3: Students will be able to provide conceptual understanding of the function of Blockchain as a method of securing distributed ledgers, how consensus on their contents is achieved, and the new applications that they enable.

CO 4: Student will able to Apply hyper ledger Fabric and Ethereum platform to implement the Blockchain Application

CO 5: Student will able to design applications based on blockchain technology for E-Governance, Land Registration, Medical Information Systems, and others

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word / sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	List out the two names of the consensus mechanism.	L1/CO2	1
Q.2	Give the name of the term used in block with definition.	L1/CO1	1
Q.3	Define merkle tree in blockchain technology.	L2/CO2	1
Q.4	State advantages of [illegible]in(any two)	L1/CO1	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answers in b[illegible] two, three Sentence).**

Q. No.	Questions	Level/CO Mapped	Marks
Q.5	Define fork and their types present in blockchain Technology.	L1/CO2	2
Q.6	State about smart contracts in blockchain technology.	L1/CO1	2
Q.7	Discuss the types of blocks(any two).	L2/CO1	2
Q.8	Distinguish between symmetric and asymmetric key cryptography.	L2/CO2	2

SECTION C

1. **Solve any two questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.9	Describe hashing technique, hash pointer, message digest with diagram.	L6/CO2	4
Q.10	Explain different types of Blockchain Technology.	L5/CO1	4
Q.11	Illustrate the following 1.Miners 2.Digital signature	L4/CO1, CO2	4